

Final ReALIGNN Result Tables and Figures

ALIGNN Reimplementation Study

May 2026

1 Scope

This document summarizes the final result tables and plots for ReALIGNN, our reimplementation and target-specific improvement of the original local ALIGNN baseline. The final result set includes formation energy and the four follow-up targets:

- `formation_energy_peratom`
- `optb88vdw_bandgap`
- `ehull`
- `mbj_bandgap`
- `bulk_modulus_kv`

No additional training was required for this final figure-generation step. The plots and tables were generated from the completed same-split comparison results documented in the target reports under `report/` and from the saved prediction artifacts under `results/logs/`.

2 Final Tables

The machine-readable table is available at:

`report/tables/final_realignn_results.csv`

The LaTeX table source is available at:

`report/tables/final_realignn_results.tex`

Main Test Metrics							
Target	Model	Orig. MAE	Our MAE	MAE Gain	Orig. RMSE	Our RMSE	RMSE Gain
Form. energy	Single	0.1165	0.0923	20.8%	0.1651	0.1468	11.1%
optB88 gap	Single	0.2681	0.2349	12.4%	0.6487	0.6092	6.1%
Ehull	Single	0.0550	0.0321	41.6%	0.0873	0.0736	15.6%
MBJ gap	Single	1.0066	0.8997	10.6%	1.6245	1.3993	13.9%
Bulk modulus	Single	14.7176	14.5057	1.4%	24.1459	24.5411	-1.6%
Bulk modulus	Ensemble	14.7176	14.3275	2.7%	24.1459	23.8082	1.4%

Tail Error and Per-Sample Wins

Target	Model	Orig. P95	Our P95	P95 Gain	Wins/Losses
Form. energy	Single	0.3320	0.2851	14.1%	606/394
optB88 gap	Single	1.4761	1.1939	19.1%	717/283
Ehull	Single	0.1766	0.1535	13.1%	788/212
MBJ gap	Single	3.3123	2.9459	11.1%	496/504
Bulk modulus	Single	53.3365	46.4395	12.9%	515/485
Bulk modulus	Ensemble	53.3365	44.9687	15.7%	—

3 Figures

The final figures were exported as both `.png` and `.pdf` files under `report/figures/`.

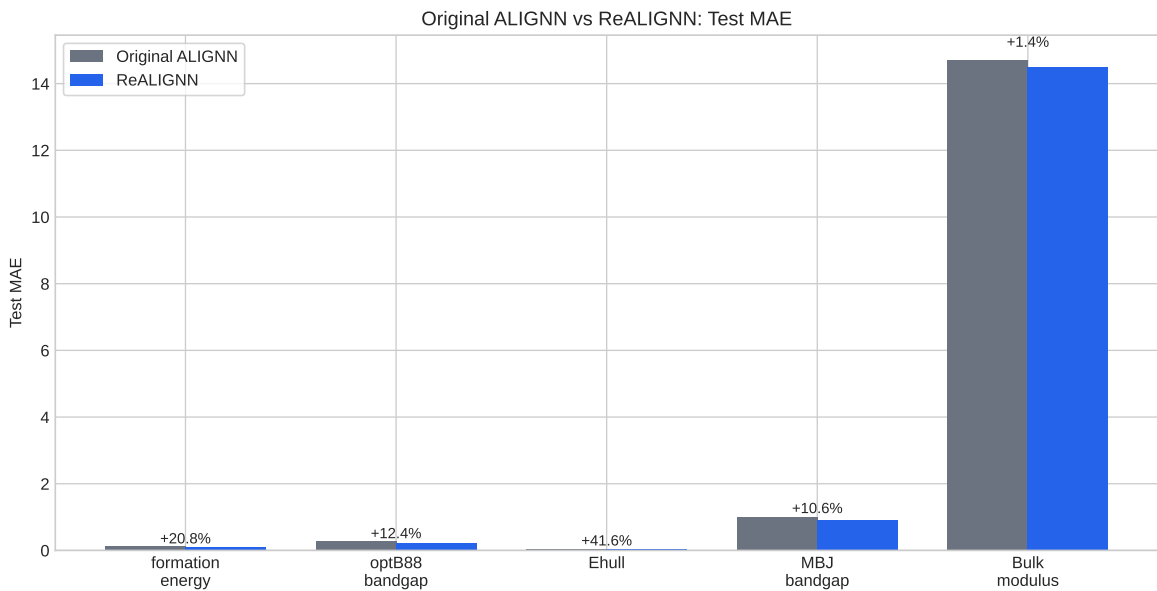


Figure 1: Same-split test MAE comparison between the original ALIGNN baseline and ReALIGNN.

4 Interpretation

ReALIGNN improves the original same-split baseline for formation energy, `optb88vdw_bandgap`, `ehull`, and `mbj_bandgap` across MAE, RMSE, and P95 absolute error. For `bulk_modulus_kv`, the best strict single model improves MAE and P95 error but has slightly worse RMSE due to a small number of hard high-error structures. The validation-selected ensemble improves bulk-modulus MAE, RMSE, and P95 simultaneously, but is reported separately because it combines multiple trained ReALIGNN checkpoints.

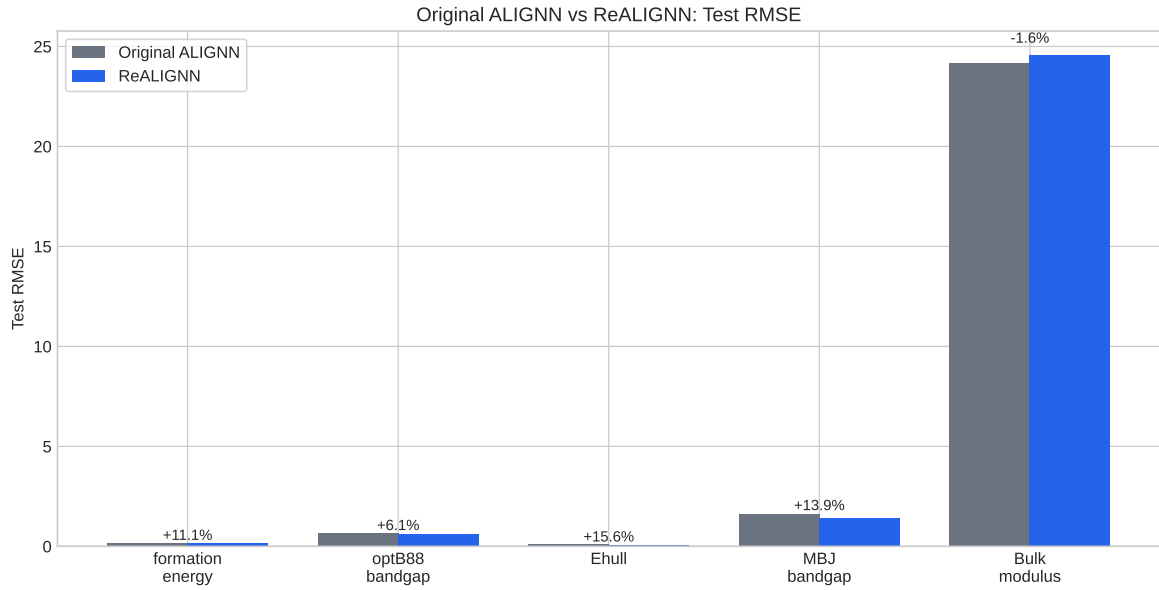


Figure 2: Same-split test RMSE comparison for strict single-model ReALIGNN runs. The strict single-model bulk-modulus run improves MAE and P95 error but not RMSE.

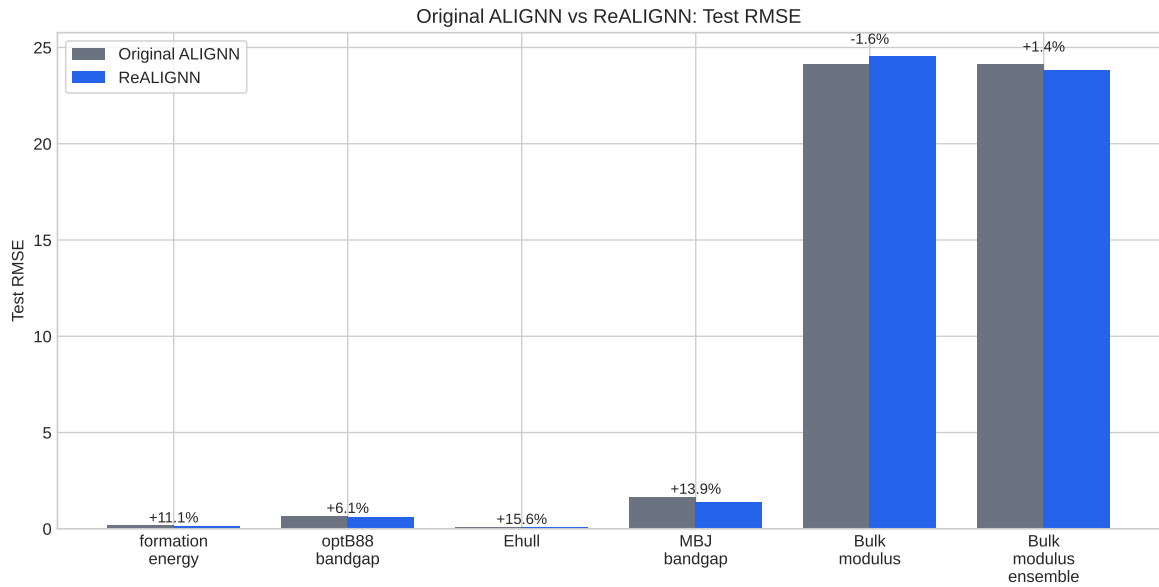


Figure 3: RMSE comparison including the validation-selected `bulk_modulus_kv` ensemble. This ensemble is listed separately from strict single-model results.

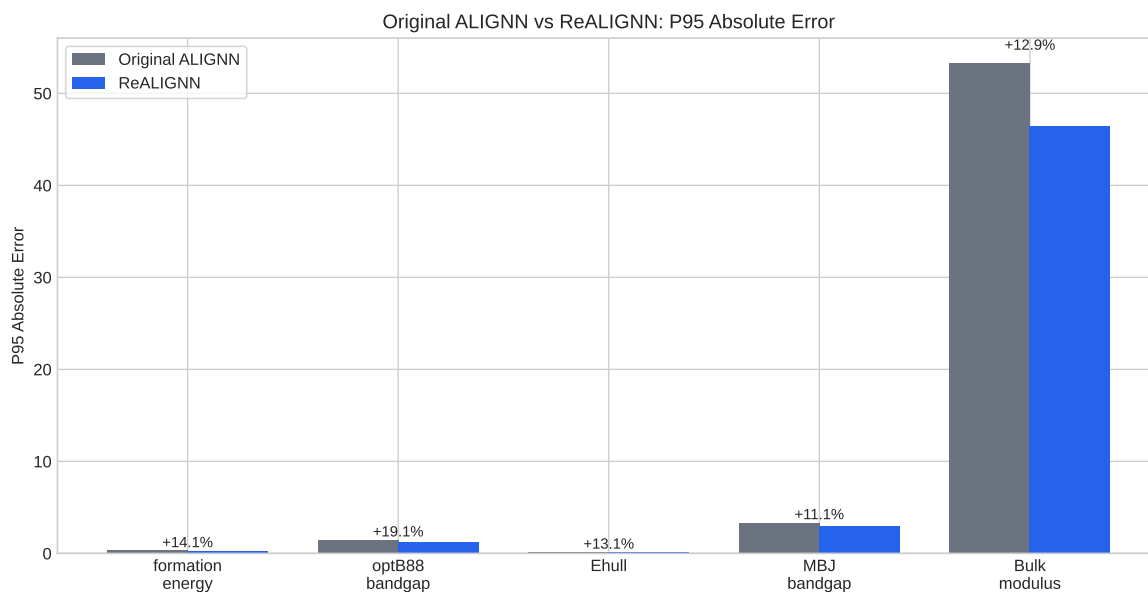


Figure 4: P95 absolute error comparison. ReALIGNN reduces tail error for all strict single-model targets shown.

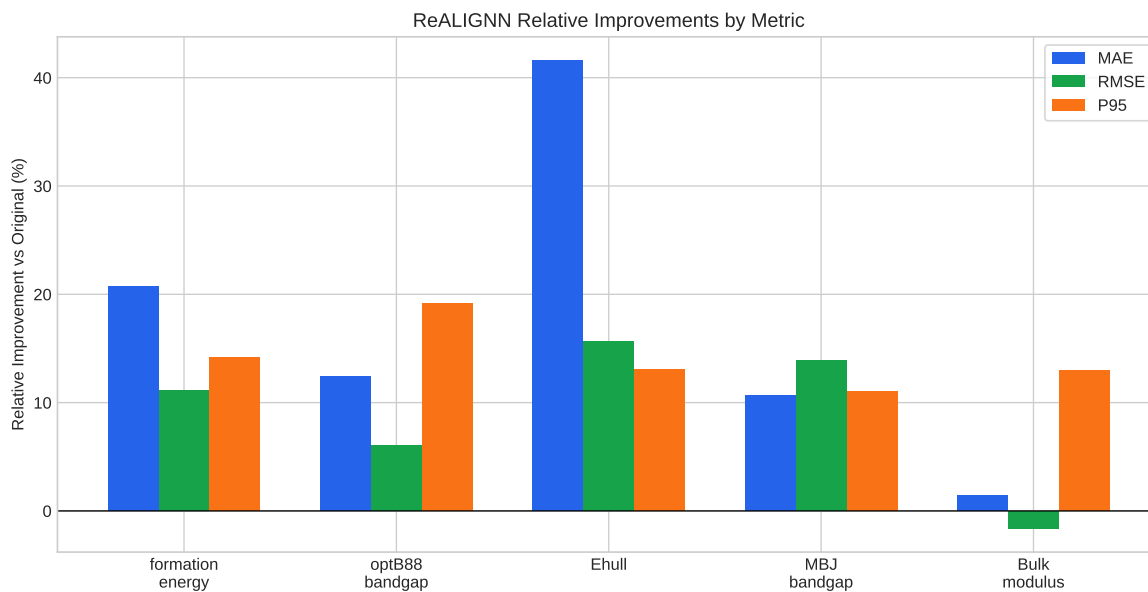


Figure 5: Relative improvement percentages for MAE, RMSE, and P95 absolute error.

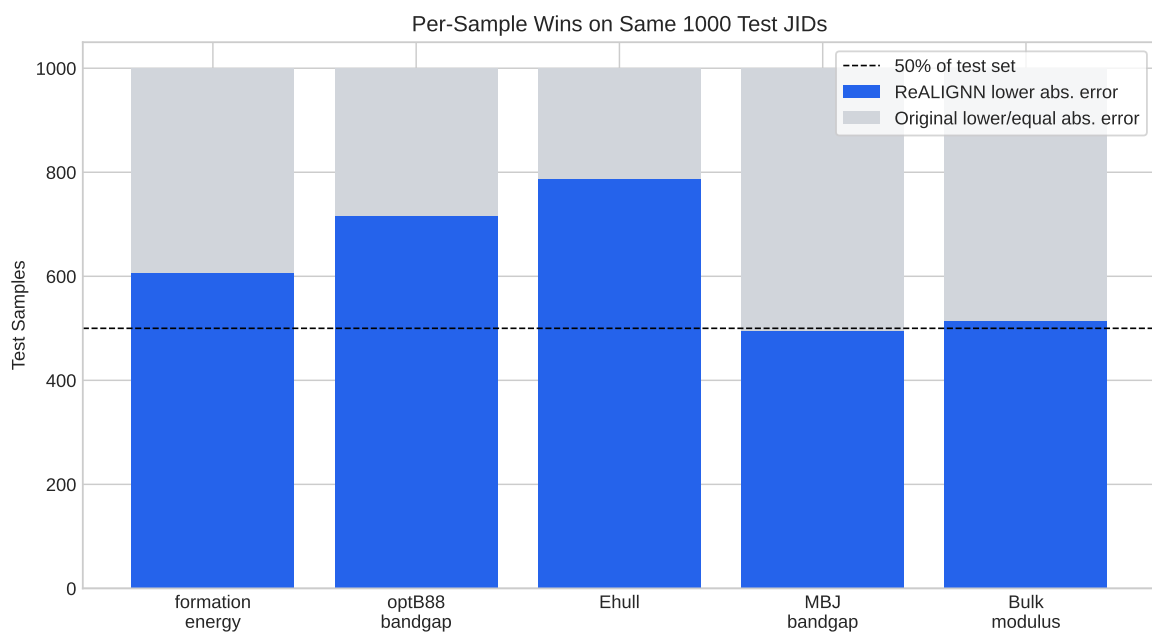


Figure 6: Per-sample win/loss counts on the exact same 1000 test JIDs for strict single-model runs.